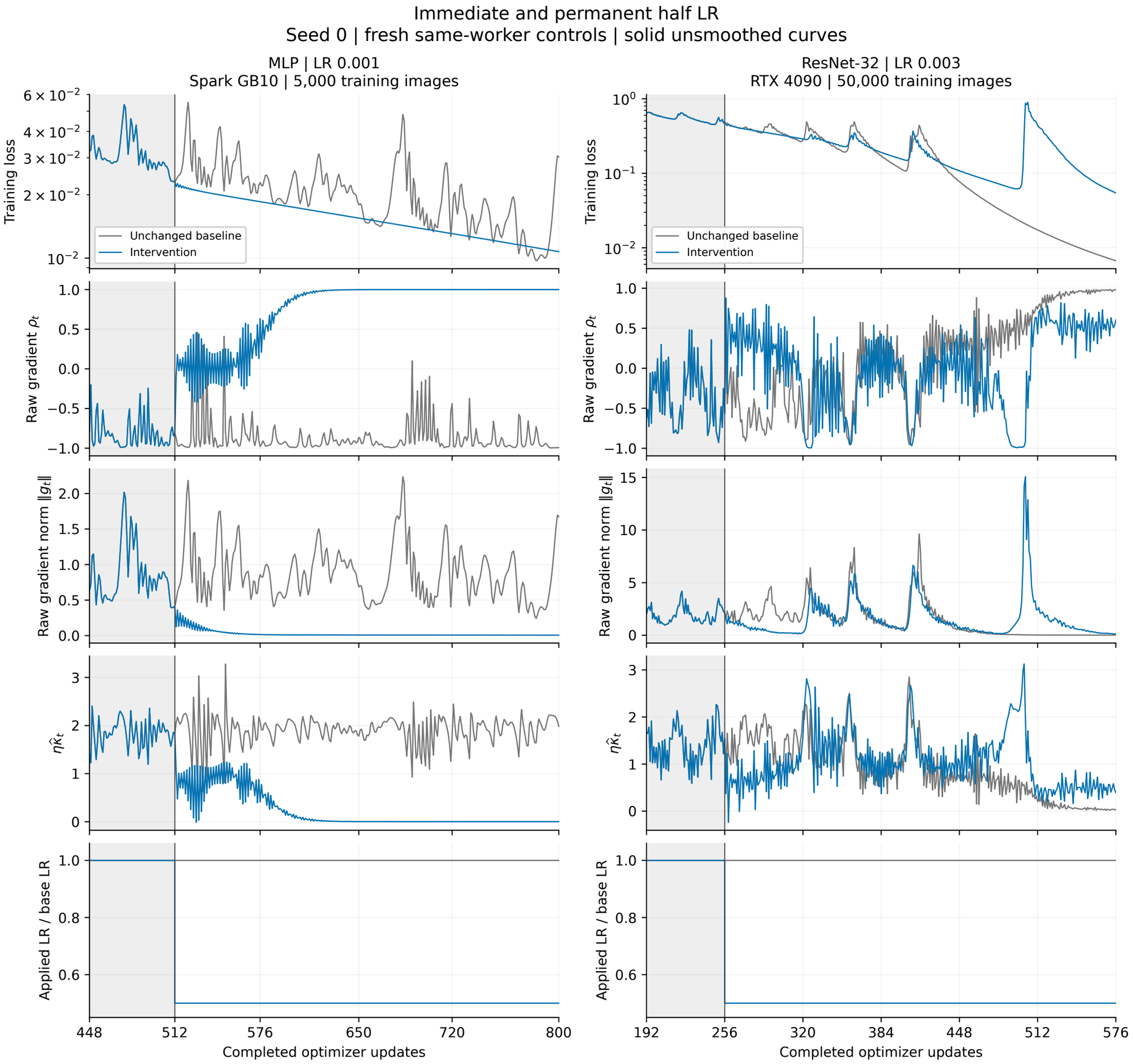

Immediate and permanent half LR
Seed 0 | fresh same-worker controls | solid unsmoothed curves
MLP | LR 0.001
Spark GB10 | 5,000 training images
ResNet-32 | LR 0.003
RTX 4090 | 50,000 training images
Training loss
Unchanged baseline
Intervention
Raw gradient ρ_t
Raw gradient norm $\|g_t\|$
$\hat{\eta}\hat{\kappa}_t$
Applied LR / base LR
Completed optimizer updates